

KnowLedge System

A self-hosted RAG stack that turns a document pool into a question-answering service — remote, multi-user, and access-controlled.

Architecture **overview** Deployment **single Linux host + GPU**

Stack **FastAPI** · **Qdrant** · **Postgres** · **BGE-M3** · **Claude**

WHY THIS EXISTS

Point it at a document pool. Get grounded answers back.

Technical specs, manuals, and compliance documents accumulate faster than anyone can search them by hand. This system parses, embeds, and indexes that pool locally, then answers questions against it through Claude — citing exactly which document and passage backed each claim.

9

documents indexed

1,462

chunks embedded

100%local embedding —
no API cost**2**ways to answer a
question

END TO END

Five stages, one request path



STAGE 01 – LOCAL, NO API COST

From raw file to searchable chunk

PARSE & CHUNK

Paragraph-aware splitting

- pypdf / python-docx / BeautifulSoup per file type
- ~1,500 characters per chunk, 200-character overlap
- Long paragraphs hard-split; short ones packed together

EMBED

**BAAI/bge-m3,
on the RTX 5090**

- Dense + sparse vectors in one pass
— strong on mixed zh/en technical text
- Cross-encoder reranker (bge-reranker-v2-m3) refines results later
- Model weights cached after first run
— no repeat downloads

CHOSEN PER UPLOAD

Two paths to the same column

Full-document questions need a structured extract of the whole document, not just top-k chunks. Generating it can run automatically against the metered API, or be deferred and run through a Claude Code session instead — same result, different billing.

AUTOMATIC

Runs at upload, unattended

1. Anthropic API call, sectioned for large docs
2. Blocks the pipeline until done or failed
3. Billed to ANTHROPIC_API_KEY, per document

CLAUDE CODE

Runs when you ask

1. `export_doc_text.py` dumps the raw text
2. `claude -p` generates the extract, sectioned the same way
3. `set_summary.py` writes it back — billed to the Code subscription

TWO RETRIEVAL MODES

Search the pool, or read one document whole

SIMILARITY SEARCH – DEFAULT

Best for open questions

1. Embed the query, hybrid search
Qdrant (dense + sparse, RRF fusion)
2. Rerank top 100 candidates down to 30
3. Claude answers from those chunks, citing filename + chunk

FULL-DOCUMENT – SCOPE-SELECTED

Best for "list all X"

1. Skips search entirely — top-k can't be exhaustive by construction
2. Uses the pre-computed summary as context (falls back to raw chunks if none exists)
3. 8,192-token answer budget — enough for genuinely long enumerations

GROUP-SCOPED, NOT PER-DOCUMENT

One filter, enforced at the vector store

Every user belongs to zero or more groups; every document is tagged with one or more groups. Retrieval is filtered by group membership directly inside Qdrant, so it stays fast at any corpus size. Admins bypass the filter and see everything.

JWT auth

Stateless session tokens, 8h expiry

Groups

Unit of access — team, project, clearance

Admin panel

Create users, edit groups, toggle admin/active

Self-lockout guard

Can't deactivate or demote your own account

`DOCKER-COMPOSE.YML`

Five services, one host

postgres

Users, groups, document
metadata, summaries

qdrant

Hybrid vector search, group-
filtered

backend

FastAPI · BGE-M3 · reranker
— GPU-passthrough

frontend

Static chat + document +
admin UI

nginx

Reverse proxy, TLS-ready,
VPN-friendly

Reachable over LAN/VPN as-is, or with a domain and Let's Encrypt for public HTTPS. A pair of export/import scripts package code, secrets, and data into one zip for moving the whole system to another server.

LIVE STATE

What's indexed right now

DOCUMENT	CHUNKS
NASC 3.0 Security Requirements Specification v1.1.0	178
NASC 3.0 Embedded OS Specification — Linux platform v1.1.0	336
NASC 3.0 Trusted Boot — User Manual v1.2.0	235
NASC 3.0 Specification Kit Release Note	19
Alternative Software Compliance Rules for NASC 3.0 Devices v1.4.1	11
NASC 3.2 Security Requirements Specification v1.3.1	185
NASC 3.2 Trusted Boot — User Manual v1.1.0	151
NASC 3.2 Embedded OS Specification — Linux platform v1.2.0	331
NASC 3.2 SRS — Tier 1 Addendum for Android-based devices v1.2.1	16

IN SHORT

Local where it can be, Claude where it matters

Parsing, chunking, embedding, and access control all run on your own hardware. Claude is called only where judgment is actually needed — generating a summary, or answering a question grounded in exactly the excerpts retrieved for it.

Embeddings **BAAI/bge-m3, local** Answers **Claude, cited** Access **group-scoped**

Portable **export / import scripts**